

Machine Intelligence (IL5HN_CR1)

Teaching Scheme

Course Code	Course Name	Contact Hours	Credits
IL5HN CR1	Machine Intelligence	04	04

1.1 Examination and Evaluation Scheme

Formative Assessment		Summative Assessment				Total Marks
ISE		MSE		ESE		
Marks	Duration	Marks	Duration (Hrs.)	Marks	Duration (Hrs.)	
20	CA	20	01	60	2 Hours	100

1.2 Course Objectives

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1	To introduce the basic concepts of Machine Learning and data pre-processing.
2	To develop in-depth understanding of various supervised learning techniques using regression.
3	To develop in-depth understanding of various supervised learning techniques using classification.
4	To discuss the use of trees and ensemble learning in Machine Learning.
5	To discuss learning with clustering and dimensionality reduction techniques.

1.3 Course Outcomes

The student will be able to:	
CO1	understand basic concepts of Machine Learning and data pre-processing.
CO2	apply various supervised learning techniques using regression.
CO3	apply various supervised learning techniques using classification
CO4	use trees and ensemble learning in Machine Learning
CO5	implement learning with clustering and dimensionality reduction techniques.

1.4 Course Contents

Module	Unit	Detailed Contents	Hours
		Pre-requisite: Fundamentals of Data structures, Database	
1.0		Fundamentals of Machine Intelligence	10
	1.1	Definition of Machine Learning, Basic Machine Learning terminology (Dataset, samples/instances, features and labels, Training, validation, and test sets model, hypothesis, parameters, and hyperparameters, Input and output variables, Loss/cost function, model training and inference), Comparison of Machine Learning with Artificial Intelligence and Deep Learning, Types of Machine Learning	
	1.2	Applications of Machine Learning, ML workflow (Problem formulation, Data collection, Data preprocessing, Feature engineering, Model selection, Model training, Model evaluation, Model deployment and monitoring)	
	1.3	Data Pre-Processing (Handling missing data, Handling categorical variables Data normalization and standardization, Removing noise and outliers, Data encoding, Feature scaling, Data splitting), Training Error, Generalization error, Overfitting, Underfitting, Bias- Variance trade-off.	
2.0		Supervised Learning using Regression	10
	2.1	Linear Regression, Multivariate Linear Regression, Logistic Regression Polynomial Regression	
	2.2	Performance Metrics: Errors, Squared Errors, SSE, SST, SSR, MSE, MAE, RMSE, R-2 Score, Adjusted R-2 Score	
	2.3	Optimization Metrics: Gradient Descent for parametric optimization, Penalty driven algorithms (Ridge/Lasso [L1 & L2])	
3.0		Supervised Learning using Classification	10
	3.1	Binary Classification: Logistic Regression with logit function non-linear prediction	
	3.2	Multi-Class Classification: Support Vector Machine for linear and non-linear classification, Naive Bayes probabilistic classification, K-Nearest Neighbors for classification.	
	3.3	Performance Metrics: Binary Cross Entropy, Categorical Cross Entropy, Confusion Matrix, Sensitivity, Specificity, Precision, Recall, F-measure, ROC curve	
4.0		Trees and Ensemble Learning	10
	4.1	Learning with Trees: Decision Trees, Constructing Decision Trees using Gini Index (Regression), Decision Tree Entropy, Classification and Regression Trees (CART).	
	4.2	Bagging: Bootstrap Aggregating, Random Subspace Method, Random Forest for Classification and Regression.	
	4.3	Boosting: AdaBoost, Gradient Boosting, XGBoost	
5.0		Learning with Clustering and Dimensionality Reduction	12
	5.1	Introduction to clustering with overview of distance metrics and major clustering approaches, Graph Based Clustering: Clustering with minimal spanning tree, Model based Clustering: Expectation Maximization Algorithm, Density Based Clustering: DBSCAN	
	5.2	Performance Metrics: Ground Truth Available (Adjusted Random Index, Normalized Mutual Information), No Label Available (Inertia-WCSS, Silhouette Coefficient, Davies-Bouldin Index.	

	5.3	Dimensionality Reduction Techniques, Principal Component Analysis, Linear Discriminant Analysis, Singular Valued Decomposition. Performance Metrics: Distance-Preservation Metrics (Sammons Stress), Neighborhood-Preservation Metrics (Co-ranking Matrix), Information-Theoretic and Variance Metrics (Kullback-Leibler [KL] Divergence)	
		Total Hours	52

1.5 Suggested Learning Resources

1.5.1 Textbooks

1. Machine Learning in Action, I Ed., Peter Harrington, DreamTech Press
2. Introduction to Machine Learning, IV Ed., Ethem Alpaydm, MIT Press
3. Machine Learning, I Ed., Tom M. Mitchell, McGraw Hill
4. Machine Learning : An Algorithmic Perspective, II Ed., Stephen Marsland, CRC Press

1.5.2 Reference Books

1. Data Mining Concepts and Techniques, III Ed., Han Kamber, Morgan Kaufmann Publishers
2. Data Mining Introductory and Advanced Topics, I Ed., Margaret. H. Dunham, Pearson
3. Education
4. Machine Learning : A Probabilistic Perspective, Kevin P. Murphy, The MIT Press
5. Introduction to soft computing, I Ed., Samir Roy and Chakraborty, Pearson Edition
6. Pattern Classification, II Ed., Richard Duda, Peter Hart, David G. Stork, Wiley Publications

1.5.3 Web Resources

1. Introduction to Machine Learning By Prof. Sudeshna Sarkar, IIT Kharagpur
https://onlinecourses.nptel.ac.in/noc21_cs85/preview
2. Data sets for Machine Learning algorithms:
<https://www.kaggle.com/datasets>
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